

Kevin Nguyen

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EDUCATION

University of Washington

Seattle, WA

B.S. in Computer Engineering — GPA: 3.6/4.0

Anticipated Graduation: June 2027

Relevant Coursework: Embedded Systems, Deep Learning (Computer Vision), Robotics Capstone (ROS2), Operating Systems, Distributed Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

Languages: C/C++, Python, TypeScript/JavaScript, SQL

Systems & Robotics: Linux, FreeRTOS, POSIX Threads, Concurrency, Networking (TCP/IP, UDP)

Embedded & Hardware: ESP32, CAN (socketCAN), UART, SPI, I2C, GPIO, PWM, Sensor Integration, PID Control

Software & Data: FastAPI, REST APIs, PostgreSQL, Redis, AWS (S3), Data Pipelines

Debugging & Tools: gdb, Valgrind, Logic Analyzers, HIL Testing, Git, Docker, CI/CD, ROS, OpenCV

EXPERIENCE

Incoming Software Engineer Intern

Jun 2026 – Sep 2026

Rivian & Volkswagen Group Technologies

Normal, IL

- Incoming Software Engineer Intern focused on vehicle software integration, testing, and automation workflows.

Software Engineer — Systems Division

Oct 2025 – Present

Husky Robotics, University of Washington

Seattle, WA

- Designed and implemented distributed CAN communication layer in C/C++ between Jetson (Linux) and embedded microcontrollers, enabling reliable cross-system integration.
- Redesigned message schemas and device addressing, improving determinism and reducing communication faults by **40%**.
- Developed internal tooling for telemetry, motor control, and system monitoring, improving observability across robotic subsystems.
- Built software-in-the-loop (SIL) validation tools and validated packet-level communication using socketCAN, vcan, and candump prior to hardware deployment.
- Implemented heartbeat monitoring and watchdog timers to detect unresponsive devices under **real-time constraints**.
- Debugged low-level communication and timing issues using logic analyzers and structured logging, resolving integration issues across hardware and software.

Project Lead & Developer

Jan 2025 – Jun 2025

Web Impact UW

Seattle, WA

- Successfully shipped production front-end web application serving **200+ users** of a club within 5 months, balancing feature delivery with stability and maintainability.
- Built CI/CD pipelines, automated testing, and deployment checks, reducing release failures by **25%**.

PROJECTS

Robotic Hand System | C++, Python, OpenCV, PyTorch, PyTest

Mar 2026 – Present

- Built a multi-servo robotic system integrating embedded control, real-time actuation, and vision-based input.
- Implemented hand tracking using **MediaPipe** and **OpenCV** to extract real-time hand landmarks from camera input.
- Developed and trained a machine learning model using **PyTorch** for hand gesture recognition, enabling mapping of gestures to robotic actuation.
- Designed PWM-based firmware to coordinate multi-channel servo control for synchronized motion.
- Developed Python-based testing and data collection pipelines to validate gesture classification and system reliability.

Megathread Clustering System | Python, FastAPI, PostgreSQL, Redis

Sep 2025 – Jan 2026

- Built scalable backend system to cluster related content using NLP techniques and REST APIs.
- Designed data pipeline processing **100+ items/sec** with optimized database queries and caching.

Drone Flight Control System | ESP32-S3, FreeRTOS, C/C++, Wi-Fi, UDP

Jun 2025 – Present

- Developed real-time embedded firmware for drone stabilization using FreeRTOS and PID control loops at high update rates.
- Implemented low-level drivers for IMU and motor control using I2C, SPI, and PWM interfaces.
- Designed wireless control and telemetry pipeline over UDP, enabling remote command and system monitoring.
- Built hardware-in-the-loop testing workflows to validate control logic and debug timing and sensor noise issues.